

**How do annual
subscribers & casual
customers use the
bikes differently?**

Divvy Bike Share, Chicago

Python analysis by Elodie Barthout ([Github](#))

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Analysis of bike rentals over the years 2018–2019

More than **6 million of rides** over this time period (6 295 548 entries):

93% of subscribers (*annual memberships*)

7% of casual customers (*single ride or day pass users*)

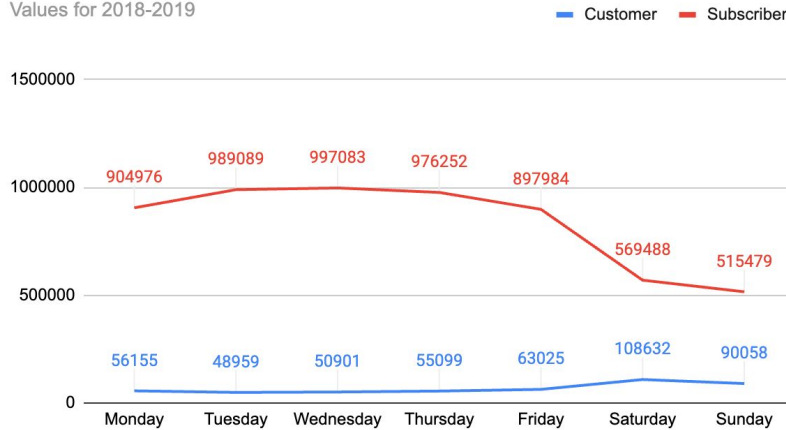
→ Majority of them being older than 30 years-old (59%)

NB: **No user IDs** → we most certainly look at the same people multiple times over this timeframe (*especially true for subscribers*)

Differences between subscribers & customers

Number of daily bike rides

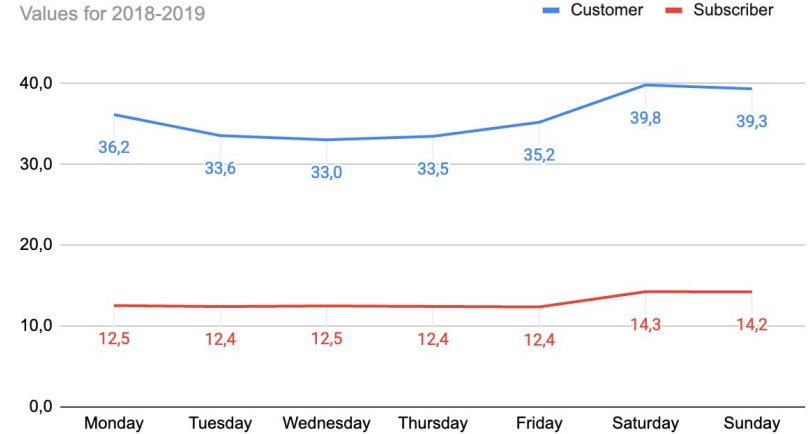
Values for 2018-2019



Subscribers mainly use the service during the week while customers seem to use it more during weekends

Average trip duration (minutes)

Values for 2018-2019



On average, customers take longer trips than subscribers (2-3 times longer), while both ride longer during weekends

Do customers take round trips?

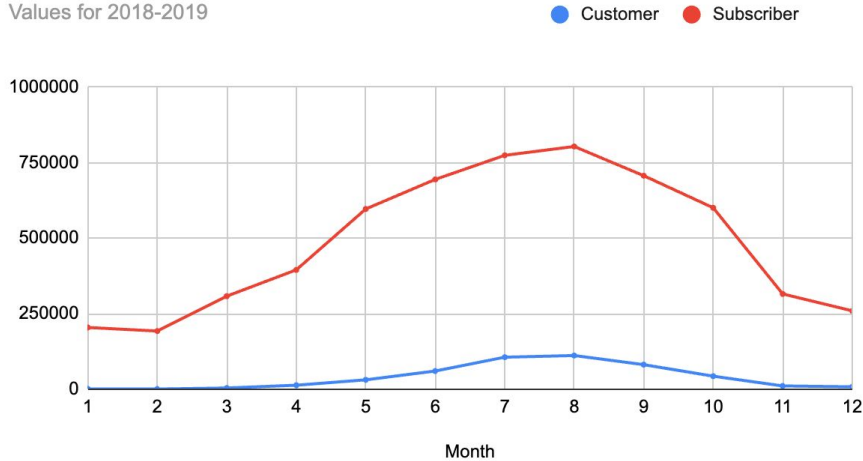
	From	To	Top
Subscriber	Canal St & Adams St	Michigan Ave & Washington St	1
	Michigan Ave & Washington St	Canal St & Adams St	2
	Michigan Ave & Washington St	Clinton St & Washington Blvd	3
	Columbus Dr & Randolph St	Clinton St & Washington Blvd	4
	Wacker Dr & Washington St	Michigan Ave & Washington St	5
Customer	Streeter Dr & Grand Ave	Streeter Dr & Grand Ave	1
	Lake Shore Dr & Monroe St	Streeter Dr & Grand Ave	2
	Lake Shore Dr & Monroe St	Lake Shore Dr & Monroe St	3
	Michigan Ave & Oak St	Michigan Ave & Oak St	4
	Streeter Dr & Grand Ave	Millennium Park	5

The most popular start & end stations for casual customers are the same stations. Further analysis showed that **8,3% of casual rides were round trips**, compared to 1,6% for subscribers. This suggests that casual customers are more likely to use bikes for recreational or sightseeing trips rather than point-to-point transportation. This is reinforced by the shore localization of the most popular stations used by casual riders.

Are there more profitable months?

Number of monthly rides

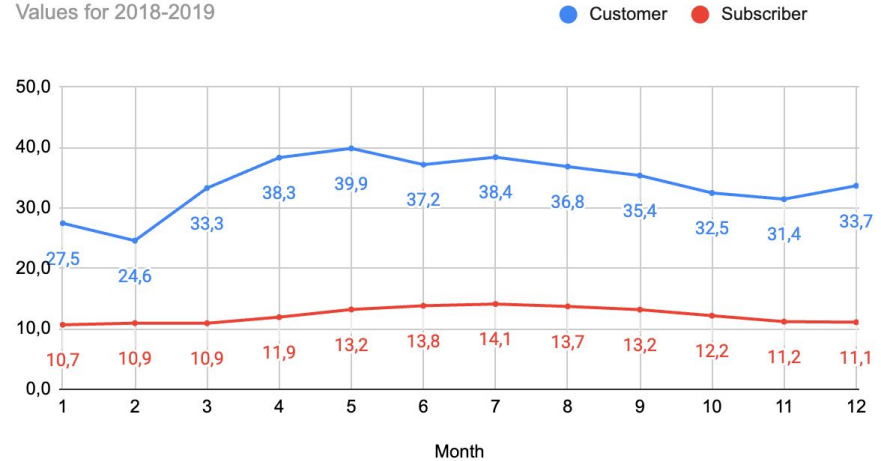
Values for 2018-2019



July & August are the months with the highest number of rides, regardless of the user type

Average trip duration (minutes)

Values for 2018-2019



Trip duration slightly increasing during Spring & Summer, and being the lowest during Winter

Conclusion

NB: No user IDs so we most probably look at the same people multiple times

- Subscribers mostly use the service during the week for an average point-to-point trip of 12 min
- Customers mostly use the service during the weekend for an average 39 min trip, mostly taking round trips around sightseeing points
- July & August are the most profitable months and represent 28% of the rides

→ Customers are probably visitors renting bikes for leisure while subscribers are probably using the bikes to commute to work